

Polymers of Intrinsic Microporosity (PIMs)

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ABSTRACT

Based on cumulative research from the past two decades, this personal perspective defines the structurally unique features of Polymers of Intrinsic Microporosity (PIMs), which result in a distinct combination of properties, including solution-processability and microporosity, providing organic materials suitable for the fabrication of membranes that perform efficient molecular sieving.

1. Introduction

Polymers of Intrinsic Microporosity (PIMs) comprise a relatively new class of macromolecule, with initial disclosure in a patent application in 2003 [1] and reported in the open literature in 2004 [2]. Subsequently, the synthesis, properties and applications of PIMs have been the subject of over 700 publications and 80 patents with research activity accelerating in recent years. Many comprehensive reviews of various aspects of PIM research have been published and the current perspective does not aim to provide yet another [3–10]. Instead, it will reflect on the results of these studies to provide greater clarity on what makes PIMs distinct from other macromolecules by identifying their unique structural features and distinguishing properties.

1.1. Microporous polymers

Micropores are defined by IUPAC as pores with a diameter less than 20 nm [11]. The formation of stable micropores within a polymer must overcome cohesive attraction between different chains (or neighbouring segments of the same chain) to avoid their collapse. Most polymers possess sufficient flexibility and rotational freedom to rearrange so as to maximise cohesive interactions and prevent microporosity. Therefore, despite the development of many types of macro- and mesoporous polymeric materials soon after Staudinger's definition of macromolecules was adopted in the 1920s [12], the first well-defined polymeric microporous materials were reported by Davankov only in the 1970s, based on extensive crosslinking of a solvent-swollen polystyrene gel [13]. The widespread commercial exploitation of Davankov's hypercrosslinked polymers (HCPs) as adsorbents is due to their attractive combination of microporosity, low cost and ease of processing into a useful bead configuration via suspension polymerization of the polystyrene precursor [13]. The inherent structure of hypercrosslinked

network polymers, within which, formally, a network of covalent bonds links all monomeric units within the polymer to each other, provides local structural stability that is well suited for maintaining microporosity. Since the disclosure of polystyrene-based HCPs, many types of microporous network polymers have been reported based on a large diversity of bond-forming reactions and precursors. Amorphous examples include the Porous Aromatic Frameworks (PAFs) [14] and Microporous Conjugated Polymers (MCPs) [15]. Of particular global interest are networks formed using reversible bond formation for which crystalline order is obtained, including the much-studied Metal-Organic-Frameworks (MOFs) [16] and the structurally related Covalent-Organic-Frameworks (COFs) [17] based on reversible covalent bond formation.

1.2. The development of PIMs

The concept of PIMs developed from the author's research on the synthesis of phthalocyanine and porphyrin based porous network polymers as potential heterogeneous catalysts [18]. In order to stop co-facial aggregation of the large planar aromatic macrocycles, fused spirobisindane (SBI) units were used to enforce a perpendicular arrangement [19,20]. These were introduced by dibenzodioxin formation using cheaply available 5,5',6,6'-tetrahydroxy-3,3',3'-tetramethyl-1,1-spirobisindane **1**, which is the readily prepared condensation product between acetone and catechol. This strategy worked well providing catalytically active porous network polymers with apparent BET surface areas (SA_{BET}) of up to 1000 m² g⁻¹ (Fig. 1a) [21].

In order to evaluate the efficiency of dibenzodioxin formation, bis-catechol **1** was reacted with commercially available 2,3,5,6-tetrafluoroterephthalonitrile **2** to form a non-network polymer (Fig. 2) by researcher Kadhum Msayib. The resulting yellow, highly fluorescent

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product proved to be a soluble polymer with a high molecular mass, as demonstrated by gel permeation chromatography (GPC) and by its formation of robust self-standing films. Surprisingly, in powder form, the polymer (eventually to be called PIM-1) demonstrated nitrogen adsorption at 77 K with significant uptake at low relative pressure suggesting that the material was predominately microporous. A description of PIM-1 and a series of related polymers, all prepared using the same dibenzodioxin formation (i.e. aromatic nucleophilic substitution), was published in a communication entitled “*Polymers of Intrinsic Microporosity (PIMs): robust, solution-processable, nanoporous materials*” [2]. The immediate potential of PIMs as membrane materials was quickly understood by Peter Budd, the author’s longstanding collaborator, who supervised a proof-of-concept study on using PIM-1 as a membrane for the separation of phenol from water using pervaporation [22].

1.3. A PIM by any other name ... would be a microporous polymer

The name *Polymer of Intrinsic Microporosity* (PIM) was originally coined by the author [2] in an attempt to make a clear distinction between the newly developed polybenzodioxin polymers, such as PIM-1, and high free volume polyacetylenes, such as poly (trimethylsilyl propyne) (PTMSP; Fig. 1c). PTMSP, first reported by Masuda et al. in 1983 [23], was the only other polymer without a network structure to be identified as a microporous material [24], a conclusion reached due to the extraordinarily high gas permeability of its freshly cast films. Aged films of PTMSP are far less gas permeable than those that are freshly prepared, eventually losing up to 90% of the initial permeability [25], suggesting that the microporosity of PTMSP is transient rather than an *intrinsic* property of the polymer. The gas permeability of cast films of PIM-1 was not measured until 2005 [26], therefore, a direct comparison of aging between films of PIM-1 and PTMSP was not available for the 2004 communication. However, nitrogen adsorption of aged powdered samples of PIM-1 did not reduce significantly over time therefore implying, at least to the author, that its microporosity was *intrinsic* to the molecular structure of these polymers rather than relying on the history of the sample. Subsequently, it has been established that gas permeability of films of PIM-1 (and all other PIMs) also reduce on aging, in common with all glassy polymers, although at a slower rate than those of PTMSP [27]. Nevertheless, the generation of microporosity by a PIM is clearly *intrinsically* related to its macromolecular structure so that the “I” in PIM is justified. The classification of PIMs as *microporous* materials initially proved controversial, with feedback at conferences being particularly critical of the use of nitrogen isotherms to calculate a

“surface area” using Brunauer-Emmett-Teller (BET) theory (Section 3.2). Similar criticisms did not appear in the literature, with the exception of scholarly analysis of the cryogenic gas adsorption of PIM-1 [28,29]. In retrospect, whilst the criticism of the widespread reporting of BET surface areas (SA_{BET}) for most classes of porous polymers is justified [30], there is no systematic reason to exclude PIMs as microporous materials using the criteria defined by IUPAC (Section 3) [11]. The acronym PIM (preferred over the initialism P. I. M.) continues to provide excellent branding for this class of polymer, perhaps benefitting from the plural “PIMs” evoking an intoxicating, British, summer drink on the patio!

1.4. Intrinsic microporosity and internal molecular free volume

It has become clear that there is a close relationship between the concept of Intrinsic Microporosity (IM), as demonstrated by PIMs, and that of Internal Molecular Free Volume (IMFV), introduced by Swager et al. [31]. Theoretical studies have shown that random arrangements of geometric shapes with concave faces fill space inefficiently [32]. The archetypal molecule with obvious concavities is triptycene (Fig. 2a), for which the concept of IMFV was defined as the unoccupied volume of the concavities relative to the total volume of the molecule “swept-out by their aromatic faces” [31]. The minimisation of IMFV was exploited for the design of triptycene-based dyes that orientate in liquid crystals, due to the rod-like molecules filling the triptycene concavities, or for aligning flexible polymer chains by their interaction with triptycene cavities [33]. In addition, Swager et al. used IMFV as a method of generating high performance dielectric materials from enhancing free volume in various classes of polymers [34]. Intrinsic Microporosity can be thought of as a macromolecular extrapolation of the molecular concept of IMFV, in which structural units that contain concavities are fused together so that larger elements of unoccupied free volume are generated from packing in the solid state (Fig. 2b). This process is best understood and visualised using chain packing simulations which has proved to be a very important tool for analyzing [35] and even predicting [36–38] the generation of IM in polymer solids (Section 3). Previously, PIMs have been defined as polymers that provide “continuous network of interconnected, intermolecular voids, which form as a direct consequence of the shape and rigidity of the component macromolecules” [39,40]. In hindsight, this appears too general and by considering both their molecular structures and properties, a more focussed definition of PIMs will be proposed in this perspective (Section 4).

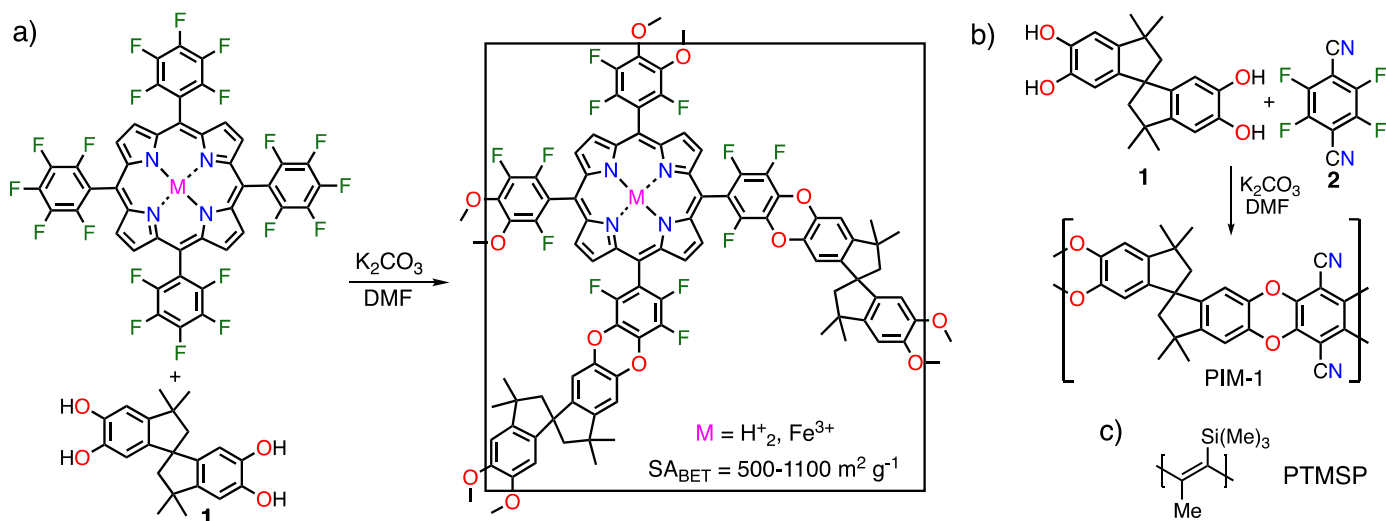


Fig. 1. (a) The synthesis of a porphyrin-based network PIM [19]. (b) The synthesis of PIM-1 [2]. (c) The structure of PTMSP.

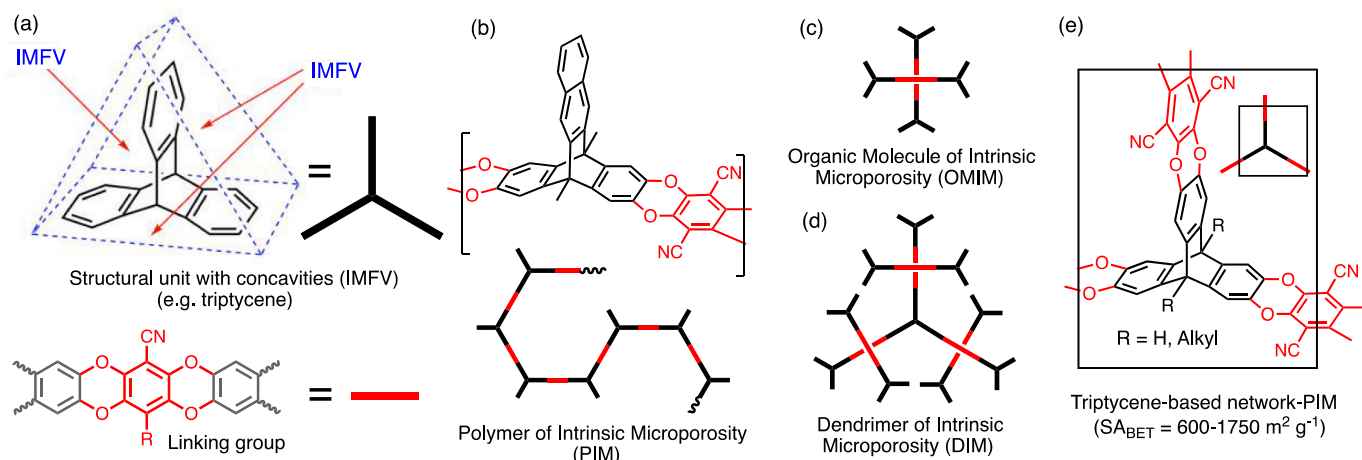


Fig. 2. (a) The concept of internal molecular free volume (IMFV) and that of intrinsic microporosity (IM) illustrated for triptycene-based PIM (b), OMIM (c), DIM (d), and Network-PIM (e).

2. Structural design rules for the generation of intrinsic microporosity

2.1. Non-network polymers

Generally, PIMs are composed of two components: firstly, a structural unit that possesses concavities and introduces a site of contortion into the polymer chain and, secondly, a linking group (e.g. derived from dibenzodioxin or imide formation) that fuses the structural units together during polymerization but prohibits rotation of one unit relative to its neighbours (Figs. 2b and 3). To use PIM-1 as an example, the structural unit is spirobisindane (SBI), derived from monomer 1, and the linking group is composed of a fused linear unit based on benzodioxin formation during polymerization with monomer 2. In addition to SBI, many different structural units have been employed for making PIMs

including spirobifluorene (SBF) [41–43], for which the fused benzo groups enhance rigidity about the spiro-carbon, and highly rigid bridged bicyclic units, such as ethanoanthracene (EA) [44–47], triptycene (Trip) [48–50], benzotriptycene (BTrip) [37,51,52] and Tröger's base (TB). TB is a particularly versatile structural unit for PIM synthesis due to its non-linear structure and ease of formation, therefore, it can be used both as the linking group during polymerization and as the structural unit by employing a TB-containing monomer [46,53,54]. When employed as a linking group, the nonlinear structure of TB acts as a site of contortion so that none is required in the diamine precursor in order to obtain a PIM (e.g. making 1,4-diamino-2,5-dimethylbenzene a suitable monomer) [53]. Due to these structural criteria, most PIMs have been prepared using step-growth polymerisations based on dibenzodioxin, TB or imide linking group formation. For the latter, it is important to restrain rotation about the imide N–C bond by the use of appropriate diamine

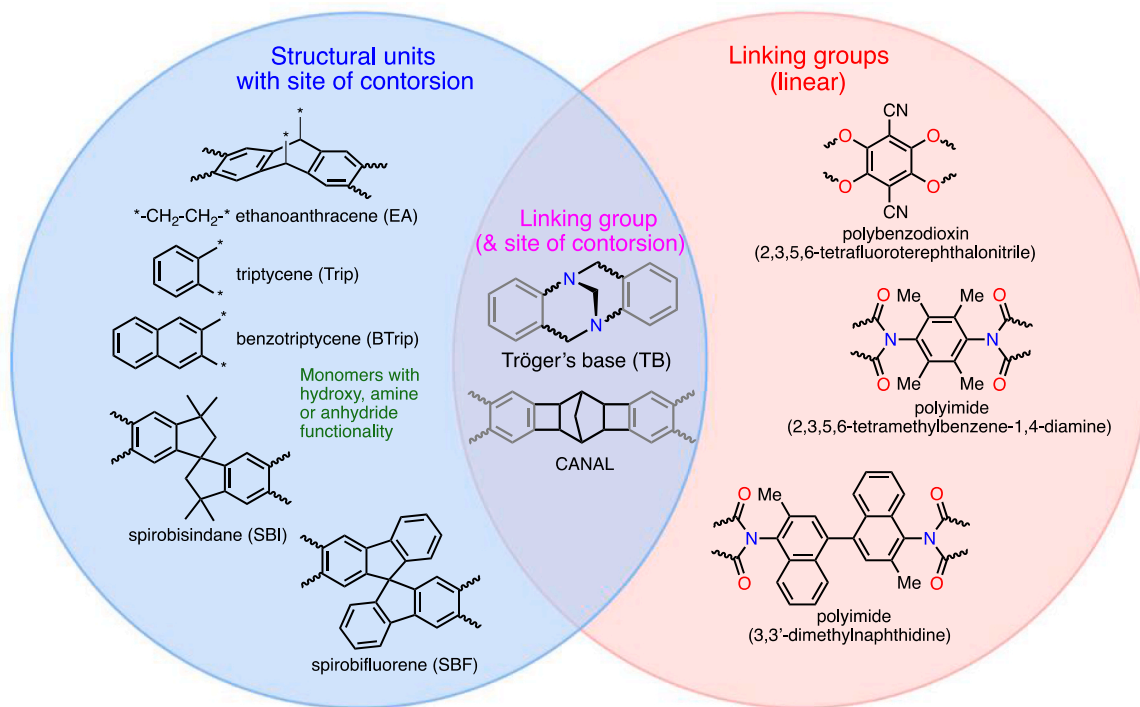


Fig. 3. PIMs are comprised of predominately fused ring chains within which structural units provide sites of contortions, to avoid linearity, with linking groups formed during the step-growth polymerisations. Tröger's base (TB) can act as both structural unit and linking group.

monomers (e.g. 2,3,5,6-tetramethylbenzene-1,4-diamine) [47,50,54–57].

An additional polymerization reaction that is fully proven to yield PIMs is based on the Catalytic Arene-Norbornene Annulation (CANAL) for which the bridged bicyclic norbornane unit provides the site of contortion within rigid fused ring structure [58–60]. It is notable that all successful PIM preparations are via a step-growth polymerization using a reaction that involves the formation of more than one covalent bond for each monomer incorporated into the polymer. It is clear that other types of polymerization reactions, such as the Diels-Alder (DA) reaction, could be used for PIM synthesis, however, despite the synthesis of some suitable DA polymer candidates [61,62], their porosity has not been investigated to date.

Post-synthetic modification of PIMs is also a useful method of tailoring properties. Most studies have focused on reactions of the nitrile group of PIM-1 to introduce carboxylic acid [63,64], amine [65], thioamide [66], tetrazole [67,68] or amidoxime [69] groups. Quaternisation of the amines in TB-based PIMs is a simple method for producing a positive charged polymer chain [70]. Most of these modifications enhance polymer cohesion and reduce microporosity but extend the structural diversity for PIMs and their potential applications.

2.2. Organic Molecules of Intrinsic Microporosity

During the early work on PIMs, it was observed that oligomeric by-products of PIM-1 gave a similar gas adsorption isotherm to that of the polymer. This observation inspired the preparation of well-defined Organic Molecules of Intrinsic Microporosity (OMIMs) [71], which involves fusing together rigid structural units with concavities, such as triptycene, using dibenzodioxin formation (Fig. 2c). Gas adsorption properties of some of these OMIMs, and that of similar oligomeric structures prepared by other groups [72], were similar to that of PIM-1 with apparent SA_{BET} values of up to 700 m² g⁻¹ [73–75]. Notably, structurally related Dendrimers of Intrinsic Microporosity (DIMs) do not show a significantly greater degree of microporosity than the OMIMs (Fig. 2d) [76]. Therefore, it can be concluded that the generation of IM is not dependent on the component molecules possessing a large molecular mass.

2.3. Network polymers

Despite the PIM concept originating from work on making porous network polymers (e.g. Figs. 1 and 2e), the identification of the role of IM in generating microporosity in polymer networks is challenging.

Many network polymers, such as the long-established HCPs, and more recently developed PAFs and network polyimides are highly microporous despite lacking the key structural criteria associated with IM – i.e. in contrast to PIMs they possess linking groups that allow full rotational freedom. Therefore, can IM be identified in network polymer and, thus, can network-PIMs be classified as a distinct subset of microporous network polymers?

For a specific network polymer, these questions might best be answered by considering the potential microporosity of an analogous non-network polymer (Fig. 4). For Davankov's HCPs, the analogous non-network polymer (i.e. polystyrene) is non-porous, therefore, there is no contribution from IM to the overall microporosity. For the "Porous Aromatic Framework" termed PAF-1, of SA_{BET} = 5500 m² g⁻¹ [14], an analogous non-network polymer could be derived from the Yamamoto coupling of di(4-bromophenyl)diphenylmethane. However such a polymer is unlikely to be microporous due to the unrestrained rotation about the biphenyl C–C bond allowing efficient packing of the chains (Fig. 4). Therefore, as for HCPs, IM does not contribute to the microporosity of the PAFs. For the porous network polyimide prepared from triaminotriptycene and 1,4,5,8-naphthalenetetracarboxylic anhydride (STPI-2) [77] a similar conclusion can be reached but in this case justified by experimental data as the equivalent non-network polyimide (TPI7) [78] is not porous. In contrast, the analogous non-network polymer to the highly microporous triptycene-based network polymer (SA_{BET} = 1750 m² g⁻¹) [79,80] is PIM-Trip (Fig. 4), which is porous (SA_{BET} = 950 m² g⁻¹), therefore, it is possible that IM contributes significantly to the microporosity of this network polymer. It has to be conceded that this analysis is subjective but it may allow the identification of the key PIM structural criteria in a network polymer and, therefore, its classification as a "network-PIM". However, this must be made with the understanding that a highly cross-linked network polymer does not require the structural features associated with a PIM to be microporous.

3. Characteristic properties of PIMs

3.1. Solubility and solution characterization

Generally, early versions of ladder-polymers were associated with very poor solubility, presumably because they were rigid rod-shaped macromolecules for which efficient packing, via chain alignment, provides strong inter-chain cohesion [81]. More recently, ladder polymer developed for use as optoelectronic materials use long alkyl chains as solubilizing groups [82]. In contrast, PIMs tend to be freely soluble in

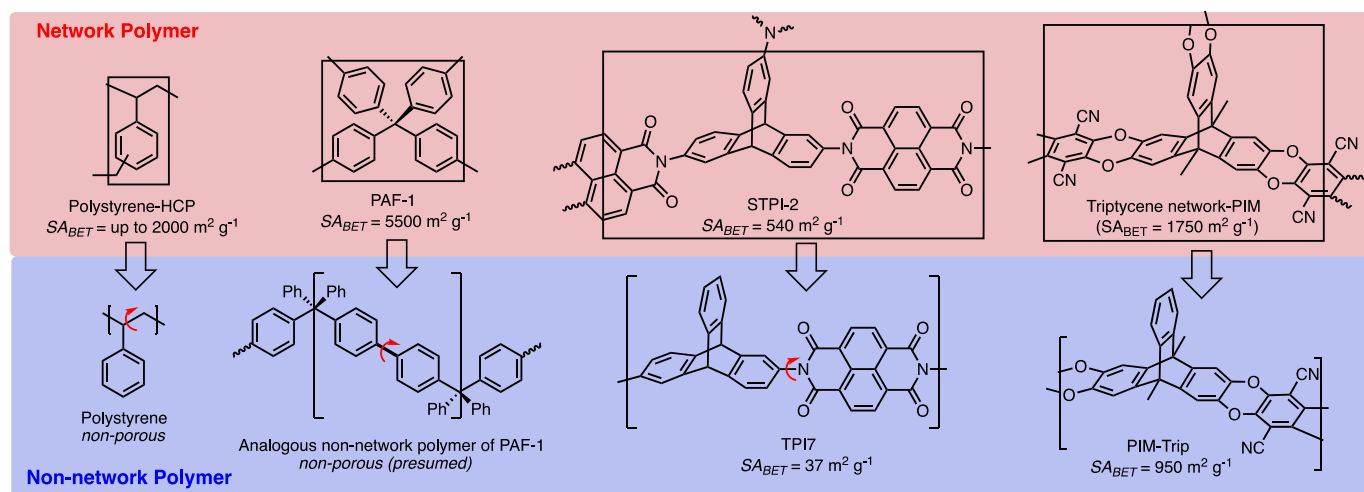


Fig. 4. Analysis of porous network polymers by comparison with analogous non-network polymers as a potential method to determine whether they can be classified as "network-PIMs". Arrows indicate bonds about which there is freedom of rotation.

organic solvents at ambient temperatures without the need for solubilizing groups. This is a key property of PIMs which has enabled their fabrication via solution processing into self-standing films for gas permeability measurements (Section 3.3), composite [83] and hollow fibre membranes [84], thin films for sensor use [85] and 3D printed objects [86]. Importantly, solubility facilitates commercial roll-to-roll fabrication (e.g. 3M's end of life indicator for their respirator cartridges) [87]. The solubility of PIMs implies that their rigid and contorted structure, which ensures inefficient packing in the solid state, also disrupts the extent of cohesive interactions by reducing the area of inter-chain contact. In addition, the IM allows access of solvent into the polymer solid to aid with dissolution. Many PIMs (e.g. PIM-1) are soluble in common low-boiling solvents (e.g. tetrahydrofuran or chloroform) but some (e.g. PIM-BTrip) require the use of high boiling aromatic solvents such as quinoline. It is notable that the rigid fused structure of PIMs means that they possess some of the highest glass transition temperatures (>350 °C) for a directly solution-processable organic material [88].

The solubility of PIMs allows their characterization by solution-based characterization methods such as NMR and GPC. NMR can be used to confirm the structure of the polymer, however, it is a distinct characteristic of the ^1H NMR spectra of PIMs that the peaks are significantly broadened as observed for rod-shaped or helical ladder polymers [89]. This phenomenon is related to greater spin-spin interactions between ^1H nuclei in polymers with fixed conformations causing shorter spin-spin relaxation times (T_2). Peak broadening appears correlated to rigidity so that the peaks of PIMs derived from bridged bicyclic structural units (e.g. PIM-EA-TB) [90] appear broader than those derived from spirobisindane (e.g. PIM-1) [91]. NMR can also indicate the degree of branching [91].

Estimation of the molecular mass of PIMs using Gel Permeation Chromatography (GPC) is also facilitated by their solubility. GPC analysis involves measuring the retention time of a polymer sample on a porous column that has been calibrated using standards of well-defined molecular mass, such as polystyrenes prepared via anionic polymerization. For the estimate to be valid, the hydrodynamic radius of the polymer chain has to be similar to that of the random-coil configuration of a polystyrene chain of the same molecular mass. For example, it is known that GPC suggests molecular mass values of rigid-rod polymers that are 50–100% higher than their actual masses [92]. Conversely, the molecular masses of OMIMs and DIMs (Fig. 2), which are highly branched PIM-like structures of known molecular mass (1500–7000 g mol $^{-1}$), are up to 40% lower than their actual values [74–76], which is consistent with the smaller values obtained for aromatic dendrimers relative to their non-branched polymer equivalents [93]. Thus, it appears that a contorted chain of a PIM may occupy a similar hydrodynamic radius as the random-coil configuration of a polystyrene chain of the same molecular mass and, therefore, GPC should provide rough estimates of the molecular masses of PIMs.

3.2. Gas adsorption analysis

Among many other physical characterization techniques [8], nitrogen (N_2) adsorption at 77 K is a standard method for measuring the porosity of solid with the resulting isotherm providing information on pore size and volume [11]. N_2 isotherms from PIMs shown significant adsorption at low relative pressure ($P/P_0 < 0.01$) indicating the presence of micropores. However, there are key differences between these and the Type I isotherms obtained from conventional microporous materials (e.g. zeolites) [11]. Firstly, for PIMs, there is continued N_2 adsorption at higher relative pressure and, secondly, there is distinct hysteresis between the adsorption and desorption isotherms. These features are consistent with the PIM swelling during N_2 adsorption as a result of the lack of a fixed framework (or flexibility for network-PIMs). Nitrogen isotherms of porous materials are used routinely to calculate the internal surface area using BET theory with the apparent SA_{BET} being routinely

reported for PIMs, despite the swelling invalidating the result [28]. Nevertheless, the value of apparent SA_{BET} has some use for making comparisons between different PIMs. Another feature of N_2 adsorption that is often observed for PIMs is the very slow uptake at low partial pressures due to the length of time it takes for the pressure of N_2 in contact with the sample to reach a stable value. This equilibration time is even longer for films than for powdered samples presumably due to the lower external surface area of the films so that N_2 molecules have further to travel to reach some adsorption sites. Nevertheless, it has been possible to obtain isotherms from very thin films and for films of PIMs with greater microporosity. These results imply a slow diffusion of N_2 into the PIM, which is consistent with gas permeability measurements (Section 3.3). The use of smaller gas probes (e.g. H_2 at 77 K or CO_2 at 273 K) allows the investigation of IM without problems associated with slow adsorption or swelling, as demonstrated by the lack of hysteresis between adsorption and desorption isotherms [28,29]. Such measurements provide evidence that the free volume within the PIMs is easily accessible to these smaller molecules (Section 3.3).

Gas adsorption data can also be used to estimate pore-size distribution (PSD) using, for example, Howarth-Kawazoe analysis, which for PIMs tends to give a smooth PSD from 0.4 to 1.2 nm with a peak value of ~ 0.5 nm (Fig. 5a) [94]. This is consistent with values obtained from the simulation of chain packing [95]. Often the default PSD analysis on nitrogen adsorption instruments use NonLocal Density Fluctuation Theory (NLDFT) with inappropriate parameter settings for organic microporous materials, resulting in artifacts such as unlikely gaps in the PSD. Parameter settings that give PSDs that are more consistent with those obtained from chain-packing simulations have been proposed by Colina et al. [96].

3.3. Gas permeability

The values for gas permeability through a film of a PIM can be used as a direct indicator of the amount of free volume present and, importantly, to assess the potential of the PIM for use as the selective layer in a gas separation membrane. Generally, the gas permeability of a gas x (P_x) through a PIM is very high relative to other types of glassy polymer, with the exception of certain polyacetylenes [97], such as PTMSP (Fig. 1c), with values that are several orders of magnitude higher than those of polymers currently used for gas separation membranes [26,37,41,43,46,49–52,54,98–106]. However, such values for all polymers are not fixed and depend greatly on the history of the film sample prior to measurement. Therefore, it is best to report gas permeabilities from films of different thicknesses, treatments and age to help other researchers understand the true potential of a new polymer. For example, PIM films recently soaked in methanol to remove any remaining casting solvent tend to demonstrate much higher permeability for all gases relative to those that have been left to age (Fig. 5). Such physical aging is a general feature of glassy polymers and involves the loss of free volume over time and is greatly accelerated for thin films [5]. In addition to permeability (P_x), selectivity (P_x/P_y or α_{xy}) for gas x over gas y is key to the performance of a membrane separation. Generally, polymers suffer from a trade-off between permeability and selectivity that has been quantified by Robeson [107,108] using double logarithmic plots of P_x versus α_{xy} for each gas pair (x/y) of interest. In 1991 Robeson defined an upper bound for each gas pair based on the data of the best performing polymers [107]. Data for PIMs based on the SBI unit (PIM-1 and PIM-7) [26] contributed to Robeson's 2008 revision of the upper bounds due to their high permeability and moderate selectivity [108]. Subsequently, the use of more rigid structural units such as SBF [41,43] and, in particular, bridged bicyclic units such as EA [46,47] and Trip [49,50,102] (Fig. 3) produced PIMs that demonstrate data well above the 2008 upper bound for O_2/N_2 , H_2/N_2 and H_2/CH_4 prompting Pinnau to propose the revision of the upper bounds for these gas pairs in 2015 [109]. In 2019, revisions for the important CO_2/CH_4 and CO_2/N_2 upper bounds were proposed using data from a series of PIM-BTrip polymers [52]. Generally, the

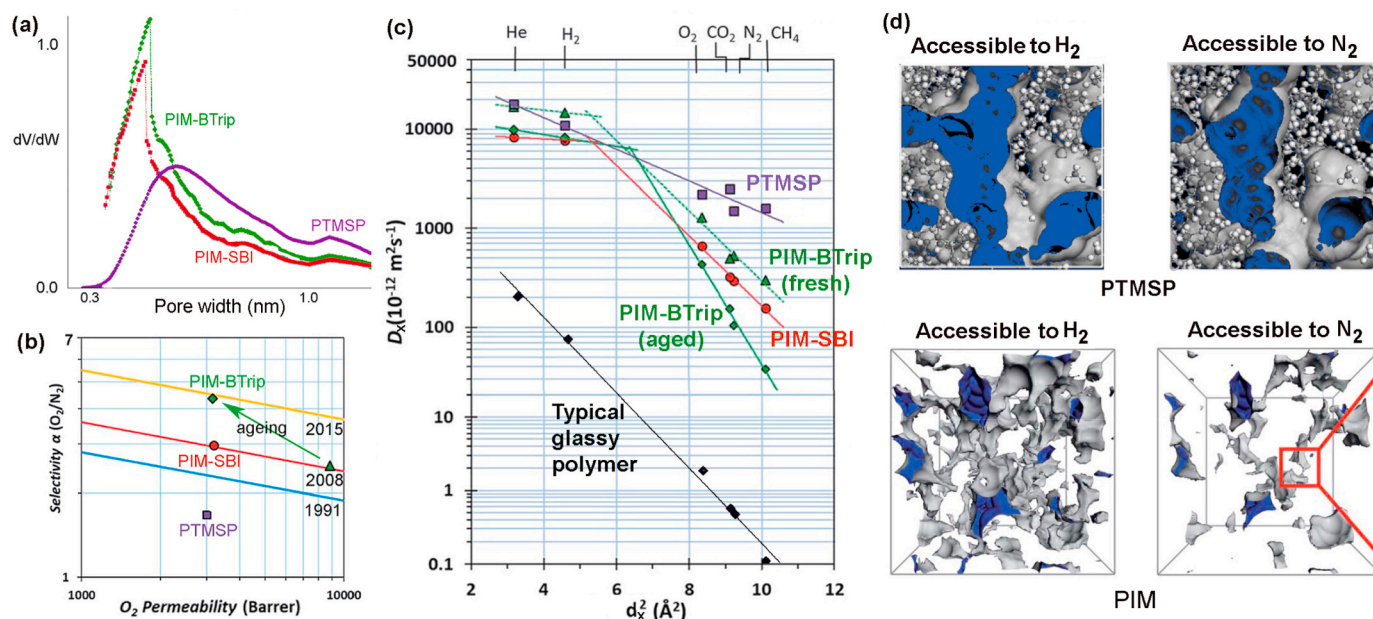


Fig. 5. Comparisons of data for a PIM based on SBI, a PIM based on BTrip (fresh and aged film) and PTMSP: (a) Pore size distributions (b) O_2/N_2 Robeson plot and (c) plot of gas diffusivity (D_x) versus the square of the gas effective diameter (d_x^2). (d) Representations of the free volume (shaded purple) that is accessible to H_2 (left) and N_2 (right) both for PTMSP [110] (top) and a PIM [101] (bottom). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

greater selectivity of PIMs relative to other high free volume polymers, such as PTMSP and related polyacetylenes, can be attributed to the distribution of the free volume (i.e. PSD) as determined directly by gas adsorption (Fig. 5a) [94] and by simulations of chain packing (Fig. 5d) [110]. For PTMSP, the free volume elements are larger and accessible to both smaller gas molecules such as He and H_2 and larger ones such as N_2 and CH_4 . In contrast, the free volume for even the most permeable PIMs is composed of smaller elements that are freely accessible only to smaller gas molecules [37,47,101].

Fuoco et al. showed that plotting the diffusivity of a gas (D_x) against the square of its kinetic diameter (d_x^2) gives information on the mechanism of transport through PIMs (Fig. 5c), as diffusion is dependent on gas molecules moving through cylindrical voids of sufficient cross-sectional area [111]. For a typical glassy polymer used for gas separation membranes, diffusivity is low due to the lack of free volume, and the difference between diffusivity for a smaller gas (e.g. D_{H_2}) may be four orders of magnitude greater than that of a larger gas (e.g. D_{CH_4}). This reflects the activated transport mechanism for which thermal motions of polymer chains are required to open a gap large enough to allow progress of the gas molecule. For ultrapermeable PTMSP, diffusivity is much larger for all gases due to the large size of its free volume elements, so that the difference between D_{H_2} and D_{CH_4} is only one order of magnitude. In addition, rapid rotations about the long Si–C bonds allow the $-Si(CH_3)_3$ groups to act as turnstiles allowing larger gas molecules to move easily from one element of free volume to another. Therefore, gas transport is via pore diffusion similar to that observed for conventional porous materials with pores of diameter >1 nm.

For PIMs, gas diffusivity is high, consistent with intrinsic microporosity providing pathways for rapid transport, but the difference between D_{H_2} and D_{CH_4} is higher than that for PTMSP by an additional order of magnitude. For the larger gases (O_2 , N_2 , CO_2 and CH_4), the gradient of the correlation between D_x and d_x^2 is similar to that of a conventional glassy polymer indicating excellent size-sieving performance, which is due to the highly rigid PIM chains restricting the opening of large voids between free volume elements. Distinctively for PIMs, the diffusivity data for the smaller gases (H_2 and He) does not lie on the same D_x versus d_x^2 correlation as that of the larger gases. This non-linearity is interpreted as the transport of H_2 and He being

predominantly via the pore diffusion mechanism, with very low activation energy for diffusivity [112]; whereas, the transport of larger gases is controlled by the activated solution-diffusion mechanism of conventional glassy polymers [111]. Hence, PIMs act like true microporous materials for small gas molecules, which experience the intrinsic microporosity as being fully accessible and interconnected, but behave more like conventional glassy polymers for larger gases (Fig. 5d). Although much less studied than gas permeability, the selective transport of ions through PIMs is of growing importance for the development of electrochemical catalysts [113–115], sensors [116] and batteries [117–121]. Similar size selectivity mechanisms to those elucidated for gases are being found for ions, with the rigidity of the PIM chain enhancing the selective transport of smaller ions over larger ions [122].

4. Conclusions

To conclude, the distinct macromolecular structure of PIMs provides organic materials with a unique combination of properties including good solubility, exceptional chain rigidity, and high free volume, which support the rapid and selective transport of small molecules. PIMs are directly solution-processable into useful forms such as membranes that can act as efficient molecular sieves. A revised definition of a PIM is proposed as a “Polymer that is composed of a fully (or predominantly) fused-ring macromolecular chain that is fixed in a 2D or 3D random coil conformation due to sites of contortion, such as bridged bicyclic or spirocyclic units, so that they pack space inefficiently in the solid state”. Beyond this definition there are (and are likely to be developed) polymers that demonstrate some PIM-like characteristics and so a more detailed summary of the distinctive features of PIM synthesis, structure and properties is given below.

4.1. Synthesis

- Aromatic monomers are reacted in step-growth polymerizations based on reactions that involve forming more than one covalent bond for each monomer incorporated into the polymer chain.

4.2. Structural

- Polymer chains are composed of aromatic rings fused together by spirocyclic, bridged bicyclic or other fused ring units so that they are fixed in a 2D or 3D random coil.
- If single bonds are present (e.g. imide C–N) sufficient steric crowding must prohibit the rotation of one aromatic ring relative to its neighbour to maintain the fixed conformation of the random coil and to ensure frustrated chain packing.
- Network polymers with these structural features may be considered as PIMs, however, not all of their porosity is likely to arise from intrinsic microporosity.

4.3. Properties

- Non-network structures are typically soluble in organic solvents, from which they will form amorphous powders by precipitation or glassy films by slow evaporation (if their molecular mass is sufficiently high).
- Polymer rigidity and fixed chain conformation results in broadened peaks in ^1H NMR spectra and very high glass transition temperatures.
- Solids possess large amounts of free volume, which is freely accessible to smaller gas molecules via pore diffusion and may be accessible to larger molecules via an activated diffusion process often accompanied by swelling.
- The rigidity of the polymer chains provides good size-selectivity for gases and ions due to the high activation energy required to open a void of sufficient size to allow larger molecules to move between elements of free volume.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

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